you spare a dime?") as the extra \$1 makes a big difference. Good times are when marginal utility is low, which in Figure 2.3 correspond to when the asset owner is wealthy and the utility function is very flat. She's already so happy ("Plenty of sunshine heading my way, zipa-dee-doo-dah, zip-a-dee-ay") that she doesn't value that extra \$1 very much.

The investor's *degree of risk aversion* governs how painful bad times are for the asset owner. Technically, risk aversion controls how fast the slope of the utility function increases as wealth approaches zero and how slowly the utility function flattens out as wealth approaches infinity. Risk aversion controls the degree of concavity in the asset owner's utility function.

Everyone wants the bird in hand. An alternative interpretation of risk aversion is that it measures just how much the investor prefers the sure thing. Consider the utility function in Figure 2.4. There are two outcomes: $X$ (low) and $Y$ (high). Suppose each of them occurs with equal probability. The two vertical lines drawn at $X$ and $Y$ correspond to the asset owner's utility at these low and high outcomes of wealth, $U(X)$ and $U(Y)$, respectively. Consider what happens at the wealth outcome $\frac{1}{2}X + \frac{1}{2}Y$, which is represented by the center vertical line. If wealth is equal to $\frac{1}{2}X + \frac{1}{2}Y$ for certain, then the asset owner's utility is given by the point on her utility curve denoted by $U(\frac{1}{2}X + \frac{1}{2}Y)$. This point is marked with the star.

Now consider the straight, diagonal line connecting $U(X)$ and $U(Y)$. If future wealth is $X$ with probability $\frac{1}{2}$ and $Y$ with probability $\frac{1}{2}$, then the expected utility

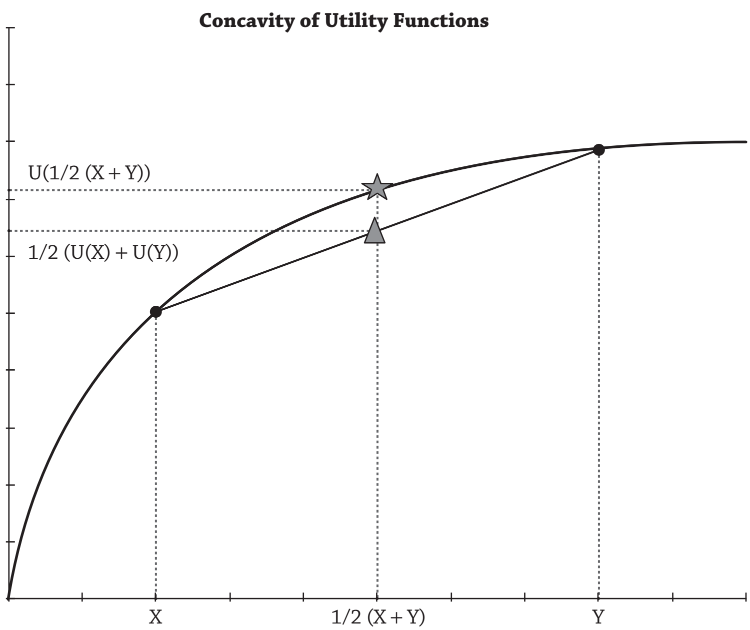


*Figure 2.4*